Annex I: Update of Debt Sustainability Analysis[1]

October 6, 2011

Based on the joint Low-Income Country Debt Sustainability Framework of the World Bank and the IMF, Nicaragua's Debt Sustainability Analysis (DSA) has been updated to reflect developments in the period following the global financial crisis. Nicaragua maintains a ***moderate risk*** *of external public debt distress, with vulnerabilities arising from worsening financing terms and a historical scenario that indicates significant risks should growth and the envisaged external adjustment not materialize. Total external debt (public and private) continues to appear sustainable in the medium term. Nicaragua was recently reclassified as a medium performer.*

I. Assumptions and Developments in Total Debt

1. **Assumptions**. Key macroeconomic assumptions were revised in the context of the seventh review of Nicaragua's ECF taking into account developments through mid-2011. Over the medium term, the baseline scenario envisages an external adjustment driven by a diversification of energy consumption away from imported oil to domestically generated renewable energy and continued fiscal prudence. Box 1 summarizes the key assumptions.

2. **External debt restructuring.** After reaching the completion point in 2004, Nicaragua received substantial debt relief under the HIPC and MDRI initiatives. Some progress has been made since then in formalizing debt relief from non-Paris Club creditors and in restructuring private commercial debt. At end-2010, the debt to non-Paris Club bilateral creditors pending relief stood at US$1.6 billion (24 percent of GDP and 40 percent of outstanding external debt).

II. External Public Debt Sustainability

3. **Under the baseline scenario all external public debt indicators remain below the policy-dependent thresholds**. Nicaragua was recently reclassified as a medium performer in terms of policies and institutional quality,[2] which lowers the debt-burden thresholds relevant for the DSA.[3] Notwithstanding this change, all external debt indicators remain below the indicative

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[1] This appendix was prepared in the context of the 7th review of the ECF arrangement that took place during August–September, 2011. It updates the last full joint Bank-Fund DSA conducted in June 2010 (SM/10/156, Supplement 1). A post-mortem of the June 2010 DSA is not included given that its assumptions and conclusions are broadly unchanged.

[2] Policy and institutional quality is measured using a three-year moving average of the World Bank's Country Policy and Institutional Assessment Index (CPIA) which has fallen for two consecutive years, crossing the threshold between the medium and strong performance categories. Nicaragua's 2010 CPIA results reflect governance concerns. The country sustains strong performances on macroeconomic, fiscal and social indicators.

[3] This reclassification reduces, among other things, the threshold for the PV of external public debt-to-GDP from 50 to 40 percent.

risk thresholds for debt distress and are expected to improve over the 20-year projection period (Table 1 and Figure 1).

Box 1. Assumptions in the Baseline Scenario

Real economic activity has recovered and real GDP is projected to grow at 4 percent in 2011. Activity is projected to slow to 3.3 percent in 2012 owing to weak external demand, before recovering to its long-term growth rate (4 percent).

The (non-interest) external current account deficit is expected to narrow gradually, from an average of 15 percent of GDP during 2005-10, to an average deficit of about 6 percent of GDP during 2026-2031, as a result of lower oil dependency as large renewable energy projects start operating.

Non-Paris Club bilateral debt is assumed to be settled on HIPC-equivalent terms by end-2011. Such restructuring would generate debt relief of about 13 percent of GDP in 2011.

Official external assistance (loans and grants to the public sector) is expected to decline from an average of 8 percent of GDP during 2006–10 to around 1 percent during 2026–31. Net official external borrowing (in concessional terms) is projected to average 3 percent of GDP during 2011-15, falling gradually over the medium term.

The deficit of the consolidated public sector (CPS) is projected to decline gradually in the coming years as the central government balance improves and the central bank reduces its operating deficit. The CPS deficit is expected to increase again in the medium term as the surpluses posted by the *Instituto Nacional de Seguridad Social* (INSS) are followed by increasing deficits (under a no reform scenario).

4. **External public debt service flows are expected to remain manageable**. External public debt service ratios are expected to remain well below the relevant thresholds reflecting the high concessionality of existing and prospective public external debt. Highly concessional borrowing is projected to decline gradually over the medium- and long-term as Nicaragua gains access to borrowing under commercial terms (see Figure 1).[4] The present value (PV) of public external debt stays at around 34 percent of GDP until 2015 and gradually declines in the following years.

5. **Standard stress tests reveal some vulnerabilities**. The "historical scenario" corresponds to a situation where the envisaged large investments in the energy sector and infrastructure do not result in a sustained increase in output growth or a decline in oil imports, and assumes that the public sector borrows to avoid an external financing gap. Under such circumstances, the non-interest current account deficit remains at its 2001–10 average level of about 15 percent of GDP, and the PV of the external-debt-to-GDP ratio increases without bound, breaching the 40 percent of GDP threshold by 2014. All debt service ratios, however, remain well below the distress thresholds. Under the standardized bound test of a “terms shock”[5] in 2012, the PV of the

[4] The temporary decline of the grant element in 2012-14 reflects the slightly lower concessionality of the large financing provided for the country’s energy program.

[5] The terms shock assumes that the interest rate on new borrowing is 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.

external-debt-to-GDP ratio marginally breaches the 40 percent of GDP threshold during most of the projection period, gradually declining by the end of the horizon. The breach under the exchange rate shock is also protracted but front-loaded, reaching 47 percent in 2012 and falling below the threshold in 2021.

6. **Private external debt at end-2010 stood at US$2.9 billion (44 percent of GDP)**.[6] Non-financial institutions account for almost one half of this debt , while the concessional financing provided by Venezuela under the auspices of the *Alternativa Bolivariana de las Americas* (ALBA), and channeled through a private cooperative, represent almost 40 percent. The remainder corresponds to debt of non-public financial institutions.

IV. Public Debt Sustainability

7. **The public debt to GDP ratio is projected to decline over the medium term**. The PV of public debt is estimated to remain stable at about 50 percent of GDP through 2014 and then to fall (to 26 percent by 2031) as a result of continued fiscal consolidation and GDP growth in line with potential. Total debt service remains below 25 percent of total revenue throughout the projection period. Continued external financing on concessional terms allows for the reduction of market-based, expensive, domestic debt. By the end of the projection horizon, pressures from the social security system begin to arise, even under the baseline assumption of fiscal discipline.

8. **Nicaragua's public domestic debt is denominated in foreign currency, and thus particularly sensitive to exchange rate shocks**. The stress tests indicate that a one-time 30 percent depreciation in 2012 would increase the PV of public debt-to-GDP and the PV of public debt-to-revenues by about 20 and 100 percentage points, respectively; the debt service-to-revenue ratio would also deteriorate significantly, reaching one-third in 2015 and falling afterwards.

V. Conclusion

9. **Nicaragua faces a moderate risk of external public debt distress**. No indicative policy-dependent threshold is breached under the baseline scenario. However, the historical scenario underscores the importance of reducing oil dependency and maintaining strong exports. Moreover, despite the assumed continued fiscal restraint, the pressures arising from the social security system start affecting the public debt dynamics noticeably towards the end of the projection period, illustrating the importance of a fundamental reform of the INSS.

10. **The authorities were in broad agreement with these conclusions and with the thrust of the analysis and the main results**. In particular, they agreed that it is imperative to strengthen the pension system and to continue transforming the country's energy matrix over the medium

[6] This estimate is considerably larger than the one reported in the previous DSA (US$1.7 billion as of end - 2009). The difference can be accounted for by improvements in the methodology and increases in the coverage for private external debt.

term. However, the authorities considered that the medium-term growth rate used in the analysis was too low. They questioned the realism of assuming that only public sector borrowing would be used to avoid an external financing gap in the “historical” scenario.

Table 1: Nicaragua: External Debt Sustainability Framework, Baseline Scenario, 2008-2031 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2008	2009	2010			2011	2012	2013	2014	2015	2016	2011-2016 Average	2021	2031	2017-2031 Average
External debt (nominal) 1/	**88.9**	**98.2**	**104.5**			**93.3**	**96.1**	**99.6**	**101.8**	**103.4**	**101.7**		**88.7**	**57.1**	
o/w public and publicly guaranteed (PPG)	56.5	60.4	60.6			45.2	46.3	47.2	47.3	46.9	46.4		42.6	32.6	
Change in external debt	-1.7	9.3	6.3			-11.3	2.8	3.5	2.2	1.5	-1.6		-3.0	-2.4	
Identified net debt-creating flows	4.7	8.7	2.0			1.4	4.7	5.6	4.2	3.5	0.2		-1.2	-1.5	
Non-interest current account deficit	**24.0**	**11.1**	**12.8**	**14.8**	**3.7**	**13.4**	**15.5**	**15.3**	**13.9**	**12.7**	**9.2**		**6.8**	**6.3**	6.5
Deficit in balance of goods and services	38.8	27.2	28.4			28.6	29.6	28.0	25.9	24.2	20.3		15.7	12.1	
Exports	46.9	46.4	55.4			64.2	67.3	67.1	67.0	66.9	66.9		66.6	66.2	
Imports	85.7	73.6	83.7			92.8	96.9	95.1	92.9	91.1	87.2		82.3	78.3	
Net current transfers (negative = inflow)	-24.2	-24.4	-21.0	-23.1	2.2	-19.6	-17.8	-17.0	-16.3	-15.6	-14.8		-11.2	-7.6	-10.2
o/w official	-5.5	-4.8	-2.8			-3.0	-2.5	-2.3	-2.1	-1.9	-1.8		-0.9	-0.2	
Other current account flows (negative = net inflow)	9.4	8.4	5.4			4.4	3.7	4.4	4.3	4.1	3.7		2.3	1.8	
Net FDI (negative = inflow)	**-9.8**	**-7.0**	**-7.8**	**-6.2**	**1.6**	**-10.5**	**-10.8**	**-8.8**	**-8.7**	**-8.6**	**-8.5**		**-8.0**	**-7.5**	-7.8
Endogenous debt dynamics 2/	**-9.4**	**4.6**	**-3.0**			**-1.5**	**0.0**	**-1.0**	**-1.0**	**-0.6**	**-0.5**		**0.0**	**-0.3**	
Contribution from nominal interest rate	0.6	2.3	2.0			2.3	2.9	2.6	2.7	3.2	3.4		3.4	1.9	
Contribution from real GDP growth	-2.2	1.3	-4.2			-3.9	-2.9	-3.6	-3.8	-3.8	-3.9		-3.5	-2.3	
Contribution from price and exchange rate changes	-7.9	0.9	-0.9			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**-6.4**	**0.6**	**4.3**			**-12.7**	**-1.9**	**-2.1**	**-2.0**	**-2.0**	**-1.8**		**-1.8**	**-0.9**	
o/w exceptional financing	-1.1	-0.6	-0.5			-13.7	-0.1	-0.1	-0.1	-0.1	-0.1		0.0	0.0	
PV of external debt 4/	...	...	77.7			81.3	83.7	86.9	88.9	90.1	88.2		74.6	44.8	
In percent of exports	...	...	140.4			126.6	124.4	129.4	132.7	134.6	131.9		111.9	67.7	
PV of PPG external debt	**...**	**...**	**33.8**			**33.2**	**33.9**	**34.5**	**34.4**	**33.7**	**32.9**		**28.4**	**20.2**	
In percent of exports	**...**	**...**	**61.1**			**51.7**	**50.4**	**51.4**	**51.4**	**50.3**	**49.2**		**42.7**	**30.6**	
In percent of government revenues	**...**	**...**	**171.6**			**157.6**	**157.8**	**160.0**	**159.8**	**155.6**	**151.6**		**128.1**	**89.7**	
Debt service-to-exports ratio (in percent)	**20.4**	**18.8**	**16.2**			**11.8**	**12.5**	**9.6**	**9.7**	**11.1**	**11.4**		**12.6**	**9.9**	
PPG debt service-to-exports ratio (in percent)	**3.9**	**4.1**	**2.7**			**2.1**	**2.3**	**2.4**	**2.6**	**3.2**	**3.3**		**3.1**	**2.7**	
PPG debt service-to-revenue ratio (in percent)	**9.6**	**10.2**	**7.5**			**6.4**	**7.3**	**7.5**	**8.2**	**9.9**	**10.2**		**9.3**	**8.0**	
Total gross financing need (Millions of U.S. dollars)	1982.6	1224.4	1308.5			1160.4	1388.8	1438.2	1397.9	1441.7	1201.6		1324.5	1608.1	
Non-interest current account deficit that stabilizes debt ratio	25.7	1.8	6.5			24.7	12.7	11.8	11.7	11.2	10.8		9.8	8.7	
Key macroeconomic assumptions															
Real GDP growth (in percent)	2.8	-1.5	4.5	2.9	2.0	4.0	3.3	4.0	4.0	4.0	4.0	3.9	4.0	4.0	4.0
GDP deflator in US dollar terms (change in percent)	9.5	-1.0	0.9	2.3	3.5	3.9	1.7	2.2	2.0	1.9	1.9	2.3	1.9	0.9	1.8
Effective interest rate (percent) 5/	0.8	2.5	2.2	1.7	0.7	2.4	3.3	2.9	2.9	3.3	3.5	3.0	4.0	3.4	3.8
Growth of exports of G&S (US dollar terms, in percent)	10.4	-3.5	25.7	13.1	10.4	25.2	10.1	6.0	5.9	5.9	5.9	9.8	5.9	5.0	5.8
Growth of imports of G&S (US dollar terms, in percent)	17.7	-16.2	20.0	10.5	11.8	19.7	9.7	4.3	3.6	4.0	1.4	7.1	5.0	5.3	5.1
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	47.3	42.0	38.3	40.6	42.3	40.4	41.8	38.2	19.3	32.8
Government revenues (excluding grants, in percent of GDP)	19.0	18.9	19.7			21.1	21.5	21.6	21.6	21.7	21.7		22.2	22.6	22.3
Aid flows (in Millions of US dollars) 7/	404.0	481.6	449.3			345.5	333.1	327.9	329.2	329.0	317.7		275.8	110.9	
o/w Grants	184.5	151.4	164.1			206.7	164.1	159.1	154.6	150.1	143.0		91.4	19.3	
o/w Concessional loans	219.5	330.2	285.2			138.8	169.0	168.8	174.6	178.9	174.7		184.4	91.7	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			4.7	4.1	3.8	3.5	3.2	2.9		1.8	0.4	1.4
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			70.5	61.4	56.7	59.3	60.9	58.6		50.9	23.4	42.9
Memorandum items:															
Nominal GDP (Millions of US dollars)	6372.2	6213.8	6551.5			7077.6	7435.4	7902.2	8383.5	8884.9	9416.3		12589.8	22185.4	
Nominal dollar GDP growth	12.5	-2.5	5.4			8.0	5.1	6.3	6.1	6.0	6.0	6.2	6.0	5.0	5.9
PV of PPG external debt (in Millions of US dollars)			2162.4			2294.2	2459.8	2660.4	2818.9	2923.0	3023.6		3495.4	4360.8	
(PVt-PVt-1)/GDPt-1 (in percent)						2.0	2.3	2.7	2.0	1.2	1.1	1.9	0.8	0.3	0.7
Gross workers' remittances (Millions of US dollars)	818.1	768.4	822.8			880.6	907.0	939.6	975.9	1012.1	1049.0		1245.2	1639.7	
PV of PPG external debt (in percent of GDP + remittances)	...	...	30.0			29.5	30.2	30.8	30.9	30.3	29.6		25.9	18.8	
PV of PPG external debt (in percent of exports + remittances)	...	...	49.8			43.3	42.6	43.7	43.8	43.0	42.2		37.2	27.5	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	2.2			1.8	2.0	2.1	2.2	2.7	2.8		2.7	2.5	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt. Historical estimates of debt have been revised (upwards) by the Central Bank of Nicaragua. The projections assume that outstanding debt to non-Paris Club bilateral creditors is settled on HIPC-equivalent terms by end-2011.

2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2a.Nicaragua: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011-2031
(In percent)

	Projections							
	2011	2012	2013	2014	2015	2016	**2021**	2031
PV of debt-to GDP ratio								
Baseline	33	34	34	34	34	33	**28**	20
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	33	36	38	41	44	49	**78**	136
A2. New public sector loans on less favorable terms in 2011-2031 2/	33	35	37	39	39	40	**41**	40
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	33	34	35	35	35	34	**29**	20
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	33	37	42	42	41	40	**34**	24
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	33	34	36	36	35	34	**29**	20
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	33	36	36	36	35	35	**30**	21
B5. Combination of B1-B4 using one-half standard deviation shocks	33	36	39	38	38	37	**32**	22
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	33	47	48	48	47	45	**39**	27
PV of debt-to-exports ratio								
Baseline	52	50	51	51	50	49	**43**	31
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	52	54	57	61	65	73	**117**	205
A2. New public sector loans on less favorable terms in 2011-2031 2/	52	52	55	58	59	59	**62**	60
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	52	49	50	50	49	48	**41**	29
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	52	58	69	69	67	66	**57**	40
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	52	49	50	50	49	48	**41**	29
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	52	53	54	54	53	52	**45**	31
B5. Combination of B1-B4 using one-half standard deviation shocks	52	54	54	54	53	52	**45**	31
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	52	49	50	50	49	48	**41**	29
PV of debt-to-revenue ratio								
Baseline	158	158	160	160	156	152	**128**	90
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	158	168	178	190	202	226	**352**	601
A2. New public sector loans on less favorable terms in 2011-2031 2/	158	162	172	179	181	183	**186**	176
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	158	158	165	164	160	155	**131**	90
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	158	170	195	193	188	183	**154**	107
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	158	158	166	166	161	157	**132**	91
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	158	166	169	168	163	159	**134**	92
B5. Combination of B1-B4 using one-half standard deviation shocks	158	170	179	178	173	169	**142**	98
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	158	218	221	221	215	209	**176**	121

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2b.Nicaragua: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011-2031 (continued)
(In percent)

	Projections							
	2011	2012	2013	2014	2015	2016	**2021**	2031
Debt service-to-exports ratio								
Baseline	2	2	2	3	3	3	**3**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	2	2	3	3	4	4	**5**	8
A2. New public sector loans on less favorable terms in 2011-2031 2/	2	2	2	3	3	4	**3**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	2	2	2	3	3	3	**3**	3
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	2	2	3	3	4	4	**4**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	2	2	2	3	3	3	**3**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	2	2	3	3	3	3	**3**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	3	3	3	3	**3**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	2	2	2	3	3	3	**3**	3
Debt service-to-revenue ratio								
Baseline	6	7	8	8	10	10	**9**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	6	7	8	9	11	12	**16**	24
A2. New public sector loans on less favorable terms in 2011-2031 2/	6	7	7	8	11	11	**10**	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	6	7	8	9	10	11	**10**	8
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	6	7	8	9	11	11	**10**	10
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	6	7	8	9	11	11	**10**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	6	7	8	9	10	11	**10**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	6	7	8	9	11	11	**10**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	6	10	11	12	14	15	**13**	11
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	23	23	23	23	23	23	**23**	23

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3.Nicaragua: Public Sector Debt Sustainability Framework, Baseline Scenario, 2008-2031
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2008	2009	2010	Average 5/	Standard Deviation 5/	2011	2012	2013	2014	2015	2016	2011-16 Average	2021	2031	2017-31 Average
Public sector debt 1/	74.3	80.0	81.8			63.4	64.0	63.7	63.3	61.6	59.8		51.2	38.4	
o/w foreign-currency denominated	74.3	80.0	81.8			63.4	64.0	63.7	63.3	61.6	59.8		51.2	38.4	
Change in public sector debt	-7.3	5.7	1.8			-18.3	0.6	-0.3	-0.4	-1.7	-1.8		-1.7	-0.4	
Identified debt-creating flows	-7.8	3.9	-2.5			-18.1	-0.5	-1.6	-2.0	-1.9	-1.9		-1.5	0.5	
Primary deficit	0.1	1.4	0.1	-0.6	1.7	-0.4	0.4	0.2	0.2	-1.1	-0.4	-0.2	0.0	1.1	0.2
Revenue and grants	21.9	21.3	22.2			24.0	23.7	23.6	23.4	23.3	23.2		22.9	22.6	
of which: grants	2.9	2.4	2.5			2.9	2.2	2.0	1.8	1.7	1.5		0.7	0.1	
Primary (noninterest) expenditure	22.0	22.7	22.3			23.6	24.1	23.8	23.6	22.3	22.8		22.9	23.7	
Automatic debt dynamics	-7.6	3.6	-2.7			-4.5	-1.4	-2.2	-2.1	-0.8	-1.5		-1.4	-0.6	
Contribution from interest rate/growth differential	-2.4	2.0	-2.9			-3.2	-1.0	-1.5	-1.4	-0.4	-1.2		-1.2	-0.8	
of which: contribution from average real interest rate	-0.2	0.9	0.5			-0.1	1.1	1.0	1.0	2.0	1.2		0.8	0.7	
of which: contribution from real GDP growth	-2.2	1.1	-3.4			-3.1	-2.0	-2.5	-2.4	-2.4	-2.4		-2.0	-1.5	
Contribution from real exchange rate depreciation	-5.3	1.6	0.2			-1.3	-0.5	-0.8	-0.7	-0.4	-0.3		...	...	
Other identified debt-creating flows	-0.2	-1.1	0.2			-13.2	0.5	0.4	-0.1	-0.1	-0.1		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.3	0.7	0.6			0.6	0.5	0.5	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.5	-1.7	-0.5			-13.7	-0.1	-0.1	-0.1	-0.1	-0.1		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.5	1.8	4.4			-0.3	1.1	1.3	1.6	0.2	0.2		-0.2	-0.9	
Other Sustainability Indicators															
PV of public sector debt	...	...	55.0			51.5	51.6	51.0	50.4	48.4	46.3		37.0	26.1	
o/w foreign-currency denominated	...	...	55.0			51.5	51.6	51.0	50.4	48.4	46.3		37.0	26.1	
o/w external	...	...	33.8			33.2	33.9	34.5	34.4	33.7	32.9		28.4	20.2	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	6.0	7.5	6.6			7.5	8.9	7.9	6.7	6.5	6.0		4.3	3.8	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	247.6			214.5	217.8	216.3	215.6	207.2	199.4		161.4	115.1	
PV of public sector debt-to-revenue ratio (in percent)	…	…	279.0			244.3	240.2	236.5	234.0	223.4	213.4		166.7	115.6	
o/w external 3/	…	…	171.6			157.6	157.8	160.0	159.8	155.6	151.6		128.1	89.7	
Debt service-to-revenue and grants ratio (in percent) 4/	23.6	22.7	19.8			15.6	21.6	20.9	18.4	24.7	21.7		16.7	11.9	
Debt service-to-revenue ratio (in percent) 4/	27.2	25.7	22.3			17.8	23.8	22.8	19.9	26.7	23.2		17.2	11.9	
Primary deficit that stabilizes the debt-to-GDP ratio	7.4	-4.3	-1.8			18.0	-0.1	0.5	0.6	0.6	1.4		1.7	1.5	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	2.8	-1.5	4.5	2.9	2.0	4.0	3.3	4.0	4.0	4.0	4.0	3.9	4.0	4.0	4.0
Average nominal interest rate on forex debt (in percent)	2.0	2.2	1.8	2.3	0.6	2.1	2.7	2.6	2.6	4.6	3.5	3.0	3.2	3.3	3.4
Average real interest rate on domestic debt (in percent)	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Real exchange rate depreciation (in percent, + indicates depreciation)	-6.7	2.1	0.2	0.0	3.0	-1.7	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	15.0	3.9	6.0	7.8	3.4	9.1	6.8	7.3	7.1	7.0	7.0	7.4	7.0	7.0	7.0
Growth of real primary spending (deflated by GDP deflator, in percent)	0.1	0.0	0.0	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	47.3	42.0	38.3	40.6	42.3	40.4	41.8	38.2	19.3	...

Sources: Country authorities; and staff estimates and projections.

1/ Public debt refers to the gross debt of the Consolidated Public Sector. The projections assume that the outstanding debt to non-Paris Club bilateral creditors is settled on HIPC-equivalent terms by end-2011.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

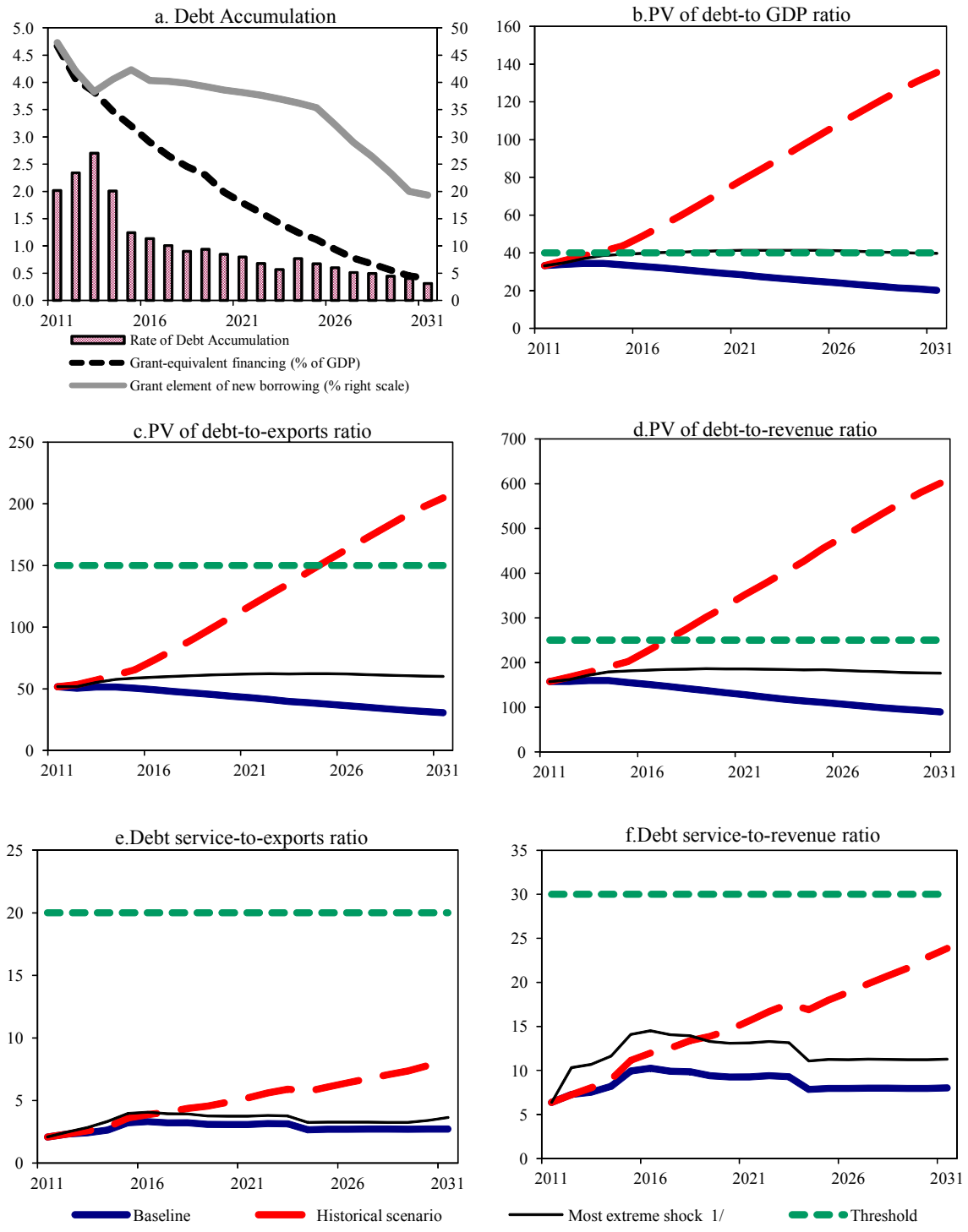
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4.Nicaragua: Sensitivity Analysis for Key Indicators of Public Debt 2011-2031

	Projections							
	2011	2012	2013	2014	2015	2016	2021	2031
PV of Debt-to-GDP Ratio								
Baseline	51	52	51	50	48	46	37	26
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	51	51	50	49	48	46	37	17
A2. Primary balance is unchanged from 2011	51	51	50	48	47	45	35	16
A3. Permanently lower GDP growth 1/	51	52	52	52	50	49	44	48
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	51	53	55	56	55	54	50	48
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	51	52	53	52	50	48	38	27
B3. Combination of B1-B2 using one half standard deviation shocks	51	52	53	53	51	50	44	39
B4. One-time 30 percent real depreciation in 2012	51	75	73	72	70	68	58	50
B5. 10 percent of GDP increase in other debt-creating flows in 2012	51	62	61	60	58	56	46	33
PV of Debt-to-Revenue Ratio 2/								
Baseline	215	218	216	216	207	199	161	115
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	215	214	211	209	205	199	159	76
A2. Primary balance is unchanged from 2011	215	214	210	207	202	194	151	73
A3. Permanently lower GDP growth 1/	215	219	219	221	215	209	190	210
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	215	225	234	239	235	232	216	212
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	215	221	223	222	214	206	168	120
B3. Combination of B1-B2 using one half standard deviation shocks	215	220	222	225	220	215	191	172
B4. One-time 30 percent real depreciation in 2012	215	317	311	308	299	291	252	219
B5. 10 percent of GDP increase in other debt-creating flows in 2012	215	260	258	257	248	240	199	147
Debt Service-to-Revenue Ratio 2/								
Baseline	16	22	21	18	25	22	17	12
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	16	22	21	18	25	22	18	11
A2. Primary balance is unchanged from 2011	16	22	21	18	24	21	16	10
A3. Permanently lower GDP growth 1/	16	22	21	19	25	22	18	17
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	16	22	22	20	26	23	19	18
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	16	22	21	19	25	22	17	12
B3. Combination of B1-B2 using one half standard deviation shocks	16	22	22	19	26	23	18	15
B4. One-time 30 percent real depreciation in 2012	16	24	26	24	33	29	24	22
B5. 10 percent of GDP increase in other debt-creating flows in 2012	16	22	23	20	26	23	18	15

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

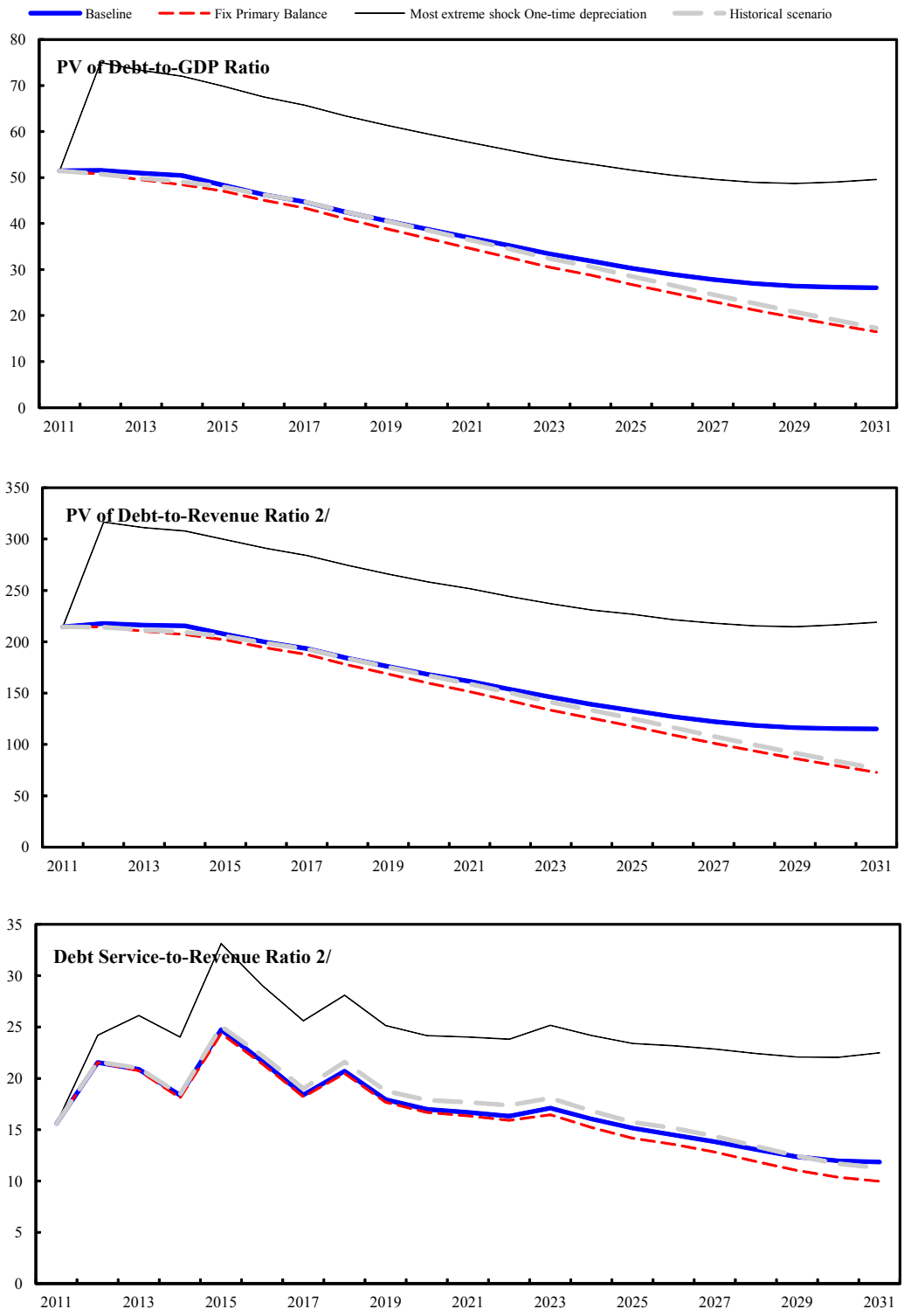
Figure 1. Nicaragua: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2011-2031 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2021. In figure b. it corresponds to a Terms shock; in c. to a Terms shock; in d. to a Terms shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2.Nicaragua: Indicators of Public Debt Under Alternative Scenarios, 2011-2031 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2021.
2/ Revenues are defined inclusive of grants.